Video Games Sales

Practice Project – Timeseries Forecasting

ABSTRACT

Throughout this report, I will demonstrate every taken step to forecast monthly sales data in order to help plan out the supply with demand for the company's video games.

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Problem Solving with Advanced Analytics-Udacity/Bertelsmann2021

Video Game Sales Forecasting

Complete each section. When you are ready, save your file as a PDF document and submit it here:
https://classroom.udacity.com/nanodegrees/nd008/parts/edd0e8e8-158f-4044-9468-3e08fd08cbf8/project

<u>Step 1</u>: Plan Your Analysis

Look at your data set and determine whether the data is appropriate to use time series models. Determine which records should be held for validation later on (250 word limit).

- The dataset at our hand is a "**Monthly Sales**" data with **No missing values**, so no need to do any data transformation/processing.
- Even though **2013** year has only **9 months**, but all are in **sequential** order with **equal spacing** from Jan to Sep

Year	Count of Month
⊞ 2008	12
⊞ 2009	12
⊞ 2010	12
⊞ 2011	12
⊞ 2012	12
⊟ 2013	9
2013-01	1
2013-02	1
2013-03	1
2013-04	1
2013-05	1
2013-06	1
2013-07	1
2013-08	1
2013-09	1
Month Grand Total	**69**

Answer the following questions to help you plan out your analysis:

1. Does the dataset meet the criteria of a time series dataset? Make sure to explore all four key characteristics of a time series data.

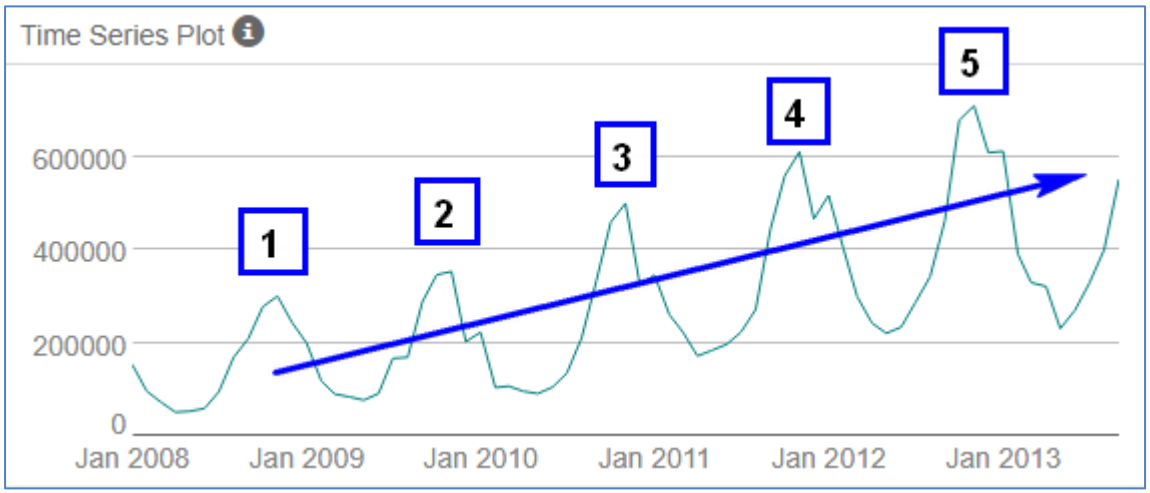
- **Yes**, the dataset **meets the criteria** of a time series dataset, <u>as it is</u>:
  - It is **time dependent** in a **continuous order** "continuous time monthly intervals".
  - Sales values exist as a **measure** at **sequential** monthly order with **equal spacing**.
  - Each tine unit (month) has **at most 1 data point** (which is the monthly sales value measure).

2. Which records should be used as the holdout sample?

- For the holdout sample (validation sample), it's good to use the **same number of periods** that we'll be forecasting for. So, **4 Records** should be used as the **holdout** sample.
- <u>I have filtered out the last 4 records</u>, **2013-06** to **2013-09**, as a holdout sample so that I can check the accuracy of my model to forecast predicted values against the actual values.

<u>Step 2</u>: Determine Trend, Seasonal, and Error components

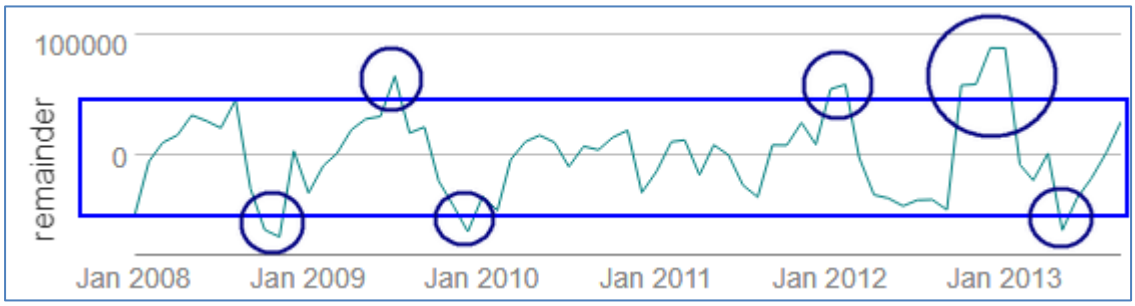
Graph the data set and decompose the time series into its three main components: trend, seasonality, and error. (250 word limit)



Answer this question:

1. What are the trend, seasonality, and error of the time series? Show how you were able to determine the components using time series plots. Include the graphs.

- **<u>Error</u>**

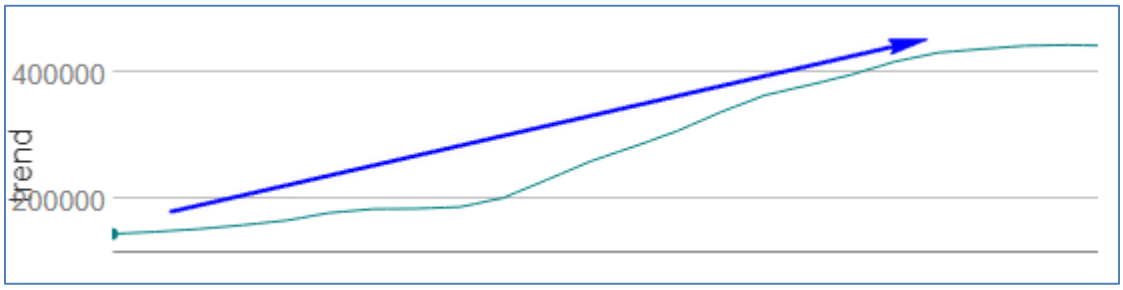


  - <u>As we know</u>,
    - If the error has **constant variance** over time (**peaks** and **valleys** are about the **same size**), then you apply it **additively**.
    - If the error is **growing** or **shrinking** over time, then you would apply the error **multiplicatively**.

  - <u>In this case as we see</u>, the error is **growing** over time, then we would apply the error **multiplicatively (M)**.

  - <u>By the way</u>, the ETS tool Auto selects **Multiplicative** also.

- **<u>Trend</u>**: Trend is clearly an **upward** trend.

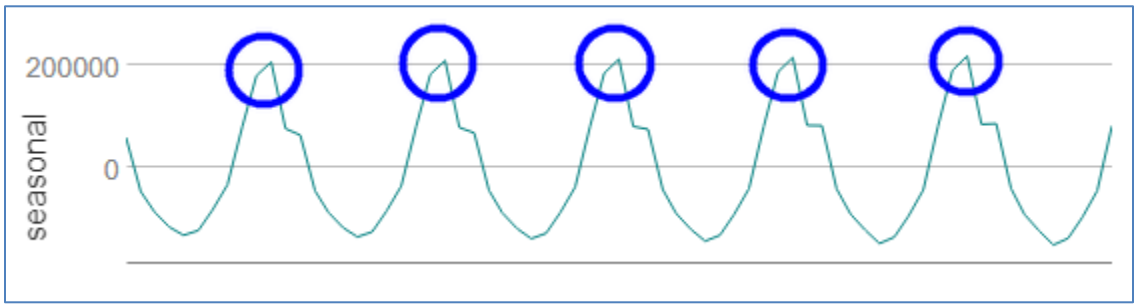


  - <u>As we know</u>,
    - If the trend is **linear**, then you apply it **additively**.
    - If the trend line **grows** or **shrinks exponentially**, then you would apply the error **multiplicatively**.

  - <u>In this case as we see</u>, Although the trend seems going up suddenly at the middle of the period, generally we can see it goes back up in constant linear increasing line, then we would apply the error **Additively (A)**.

  - <u>By the way</u>, the ETS tool Auto selects **Additive** also.

- **<u>Seasonality</u>**



  - <u>As we know</u>,
    - If the **peaks** and **valleys** for seasonality are **constant** over time, then you apply it **additively**.
    - the seasonality **grows** or **shrinks** overtime, then you would apply the error **multiplicatively**.
    - *We <u>don't have to consider</u> if the change is linear or exponential*, **only** if it is growing or shrinking over time

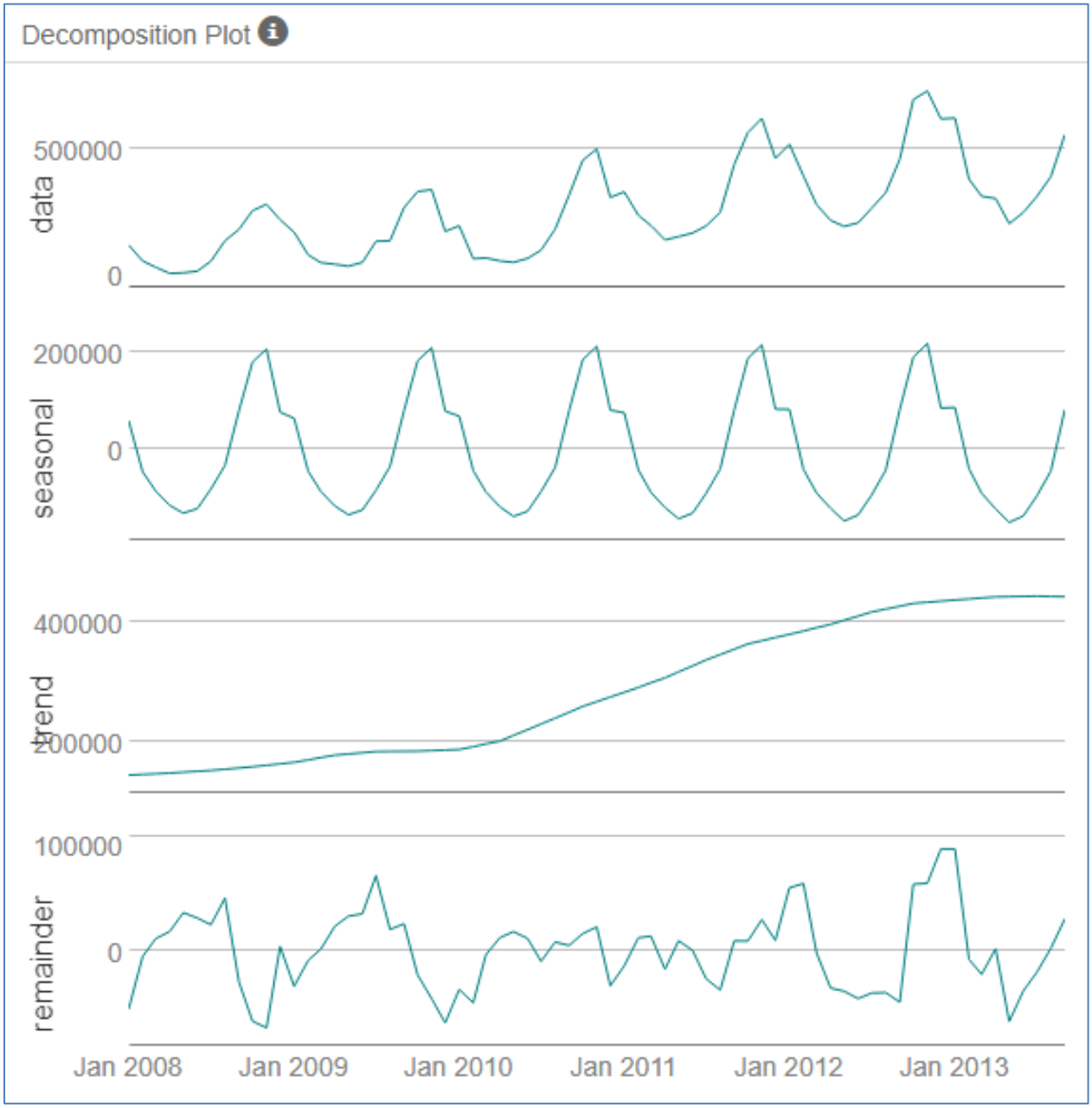
  - <u>In this case as we see</u>, Sasonality is **growing slightly over time** (the **peaks** are **increasing**, just very **slowly**), so we would apply it **Multiplicatively (M)**.

  - <u>By the way</u>, the ETS tool Auto selects **Multiplicative** also.

<u>Step 3</u>: Build your Models

Analyze your graphs and determine the appropriate measurements to apply to your ARIMA and ETS models and describe the errors for both models. (500 word limit)

- <u>As we saw above</u>, decomposition tree plot gives us the sense about the appropriate measurements to apply to **ARIMA** and **ETS** models.



Answer these questions:

1. What are the model terms for ETS? Explain why you chose those terms.

- From our decomposition plot we can obtain the necessary information to define our terms for the **ETS** model.
    - **<u>The Error</u>:** ***changes in magnitude*** as the series goes along so a **multiplicative (M)** method will be used.
    - **<u>The Trend</u>:** line exhibits **linear behavior** so we will use an **additive (A)** method.
    - **<u>The Seasonality</u>:** changes in **magnitude** each year so a **multiplicative (M)** method is necessary
- This leaves us with an **ETS(M, A, M)** model.
- <u>For a **comparison** purpose and for **checking** our readings</u>, we could also perform:
    - **ETS(M, A_d, M)** model with **Dampened** Parameter.
    - **ETS(Auto)** model.

- **ETS(M, A_d, M)** model with **Dampened** Parameter is the **best ETS** model to be chosen based on below **evaluation metrices**.
- It also the same model suggested by **ETS(Auto)** model.

a. Describe the in-sample errors. Use at least RMSE and MASE when examining results

- The in-sample error measures give us a look at **how well** our model can predict future.

ETS(M, A, M) model

Method:
ETS(M,A,M)

In-sample error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
281.3033372	40906.5798105	28212.6317577	-0.9199379	11.2444007	0.4115798	0.2971592

ETS(M, A_d, M) model with **Dampened** Parameter

```
Method:
   ETS(M,Ad,M)
```

In-sample error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
3243.4703524	31474.3668886	24188.2167878	-0.572395	10.3052041	0.3528697	0.0087402

ETS(Auto) model.

```
Method:
   ETS(M,Ad,M)
```

In-sample error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
3243.4703524	31474.3668886	24188.2167878	-0.572395	10.3052041	0.3528697	0.0087402

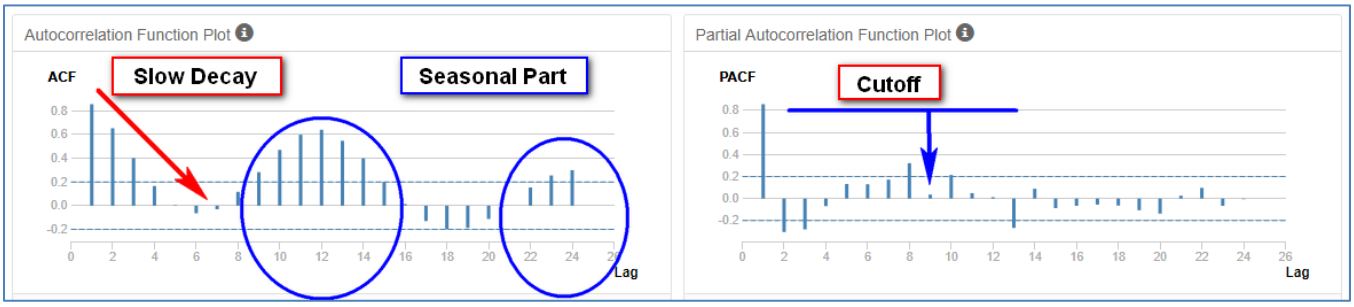
- For ETS**(M, A_d, M)** model with **Dampened** Parameter: Two key components to look at are the **RMSE**, which shows the in-sample **standard deviation**, and the **MASE** which can be used to compare **forecasts of different models**.
    - At **RMSE**, we can see that our variance is about **31000** units around the mean.
    - The **MASE** shows a fairly strong forecast at **0.35** with its value falling well below the generic **1.00**, the commonly accepted MASE threshold for model accuracy.

2. What are the model terms for ARIMA? Explain why you chose those terms. Graph the Auto-Correlation Function (ACF) and Partial Autocorrelation Function Plots (PACF) for the time series and seasonal component and use these graphs to justify choosing your model terms.

- Since there are **seasonal components** found in the time series, we will use an :
    - **ARIMA(p, d, q)(P, D,Q)m model for forecasting.**

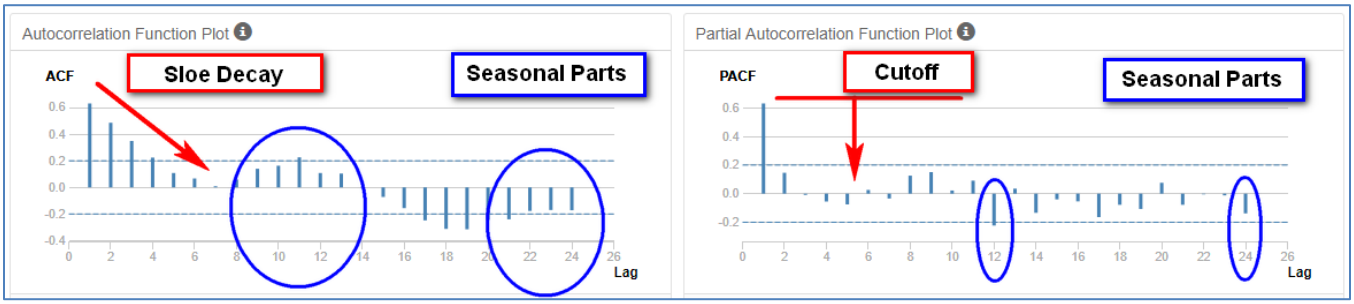
Time Series ACF and PACF:

- The **ACF** presents **slowly decaying** serial correlations towards **0** with increases at the seasonal lags. Since serial correlation is high, we will need to **seasonally difference** the series.

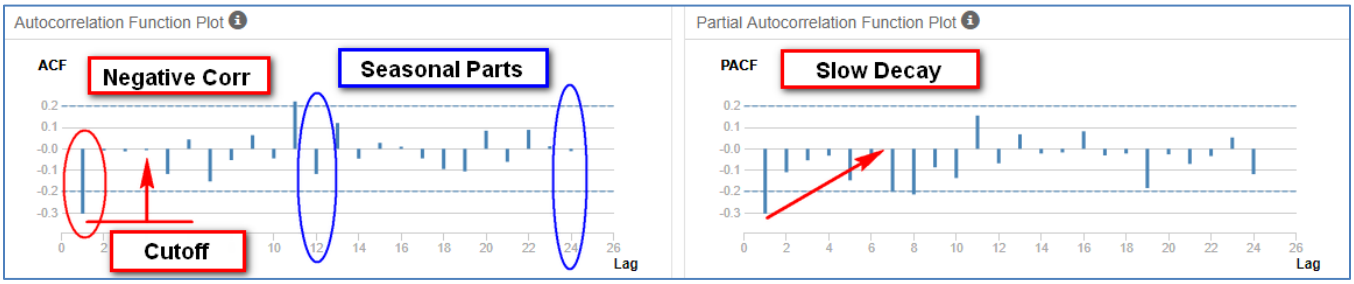


Seasonal Difference ACF and PACF:

- The seasonal difference presents **similar ACF and PACF results** as the initial plots without differencing, only **slightly less correlated**. In order to remove correlation, **we will need to difference further**.



Seasonal First Difference:



- The seasonal first difference of the series has **removed most of the significant lags** from the **ACF and PACF** so there is **no need for further differencing**.

- <u>The remaining correlation</u> can be accounted for using **autoregressive** and **moving average** terms.

- <u>We did 2 differencing</u>, 1 for the Non-Seasonal part (d) and 1 for the Seasonal part (D), so:
    - The **differencing** terms will be **d(1)** and **D(1).**

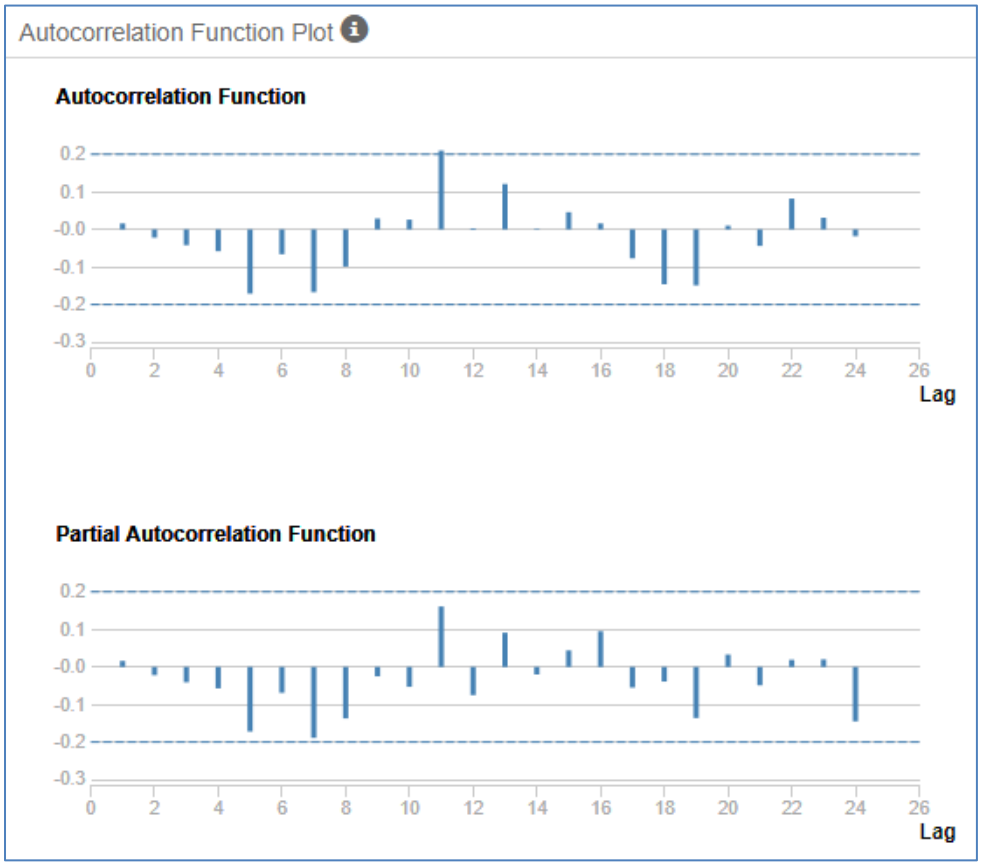
- The ACF plot shows **Cutoff towards 0**, a **strong negative correlation** at **lag 1** which is confirmed in the **PACF**, as we see **slow decay towards 0**.
    - This suggests an **MA(1) model** since there is **only 1 significant lag**.

- The **seasonal lags** (lag 12, 24, etc.) in the ACF and PACF **do not have any significant correlation** .
    - So, there will be **no need for seasonal autoregressive** or **moving average** terms.

- Therefore, the model terms for my ARIMA model are:
    - **ARIMA(0, 1, 1)(0, 1, 0)[12]**

a. Describe the in-sample errors. Use at least RMSE and MASE when examining results
b. Regraph ACF and PACF for both the Time Series and Seasonal Difference and include these graphs in your answer.

- The **ACF** and **PACF** results for the **ARIMA(0, 1, 1)(0, 1, 0)[12]** model shows **no significantly correlated lags** suggesting **no need for**



- The in-sample error measures give us a look at **how well** our model can predict future.

In-sample error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
-356.2665104	36761.5281724	24993.041976	-1.8021372	9.824411	0.3646109	0.0164145

- <u>For ETS**(M, A_d, M)** model with **Dampened** Parameter</u>: Two key components to look at are the **RMSE**, which shows the in-sample **standard deviation**, and the **MASE** which can be used to compare **forecasts of different models**.

    - At **RMSE**, we can see that our variance is about **37000** units around the mean.

    - The **MASE** shows a fairly strong forecast at **0.36** with its value falling well below the generic **1.00**, the commonly accepted MASE threshold for model accuracy.

<u>Step 4</u>: Forecast

Compare the in-sample error measurements to both models and compare error measurements for the holdout sample in your forecast. Choose the best fitting model and forecast the next four periods. (250 words limit)

Answer these questions:

1. Which model did you choose? Justify your answer by showing: in-sample error measurements and forecast error measurements against the holdout sample.

- When fitting a forecasting model, we can use a **series of identifiers** that help us choose the best model.

In-Sample error measures

- When comparing the two in-sample error measures we used, the **RMSE** and **MASE**, we see **very similar results**.
- The ETS model does have a **narrower standard deviation** but **only by a few** thousand units.
- Further investigation shows that the **MAPE** and **ME** of the **<u>ARIMA</u>** <u>model are **lower** than the **ETS**</u>. This suggests that, on average, the ARIMA model misses its forecast by a **lesser amount**.

ETS

In-sample error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
3243.4703524	31474.3668886	24188.2167878	-0.572395	10.3052041	0.3528697	0.0087402

ARIMA

In-sample error measures:

ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
-356.2665104	36761.5281724	24993.041976	-1.8021372	9.824411	0.3646109	0.0164145

Holdout Sample error measures

- When looking at the model's ability to predict the holdout sample, we see that the **ARIMA** model has **better predictive qualities** in just about every metric.

Accuracy Measures:

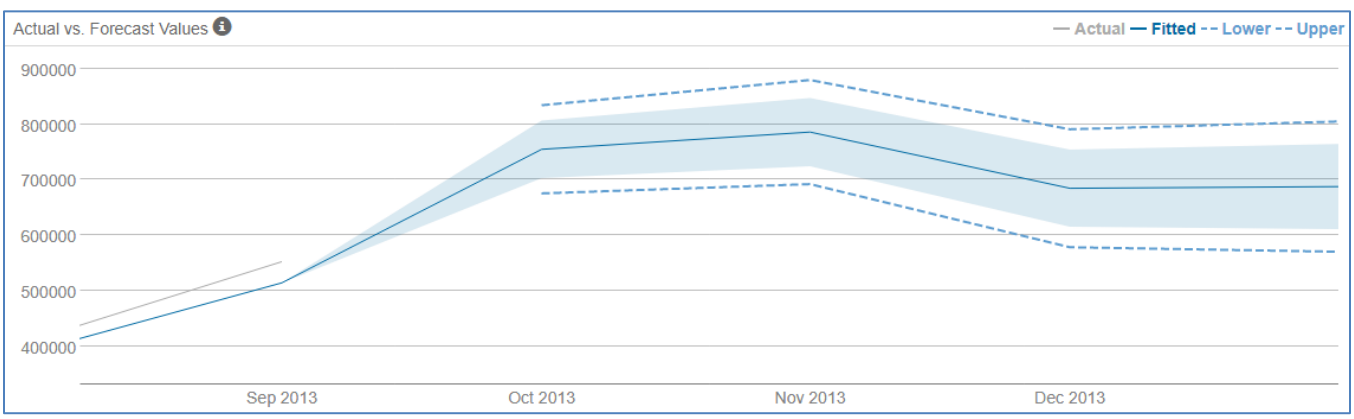
Model	ME	RMSE	MAE	MPE	MAPE	MASE
ARIMA__0__1__1___0__1__0___12__L_S_Diff	27271.52	33999.79	27271.52	6.1833	6.1833	0.4532
ETS_MAM_dampened_	-33469.61	53828.48	41542.75	-6.3476	9.3266	0.6904

2. What is the forecast for the next four periods? Graph the results using 95% and 80% confidence intervals.

- Forecast results using 95% and 80% confidence intervals:

Record	Period	Sub_Period	ARIMA_100_110_12	ARIMA_100_110_12_high_95	ARIMA_100_110_12_high_80	ARIMA_100_110_12_low_80	ARIMA_100_110_12_low_95
1	2,013	10	754,854.460048	834,046.21595	806,635.165997	703,073.754099	675,662.704146
2	2,013	11	785,854.460048	879,377.753117	847,006.054462	724,702.865635	692,331.166979
3	2,013	12	684,854.460048	790,787.828211	754,120.566407	615,588.35369	578,921.091886
4	2,014	1	687,854.460048	804,889.286634	764,379.419903	611,329.500193	570,819.633462

- Shown graphically here:



Before you submit

Please check your answers against the requirements of the project dictated by the rubric. Reviewers will use this rubric to grade your project.

Thank You